LuoLinguaAI Phase Two — Project Summary

Purpose: Comprehensive overview of Phase Two deliverables, required services, cost estimates (KES 130/USD), AI data readiness, and timeline. All prices exclude labour.

WHAT PHASE TWO WILL DELIVER

- An Android app and an iPhone app connected to the current LuoLinguaAI system.
- Lessons and approved content for the dictionary, proverbs, riddles, selected stories, songs and cultural items.
- Search, quizzes, saved words and simple learning progress.
- Downloaded lessons that can be read without internet. Progress made offline should upload when the phone reconnects.
- AI chat that uses approved project information and shows which sources it used.
- Phone notifications, email account messages, and optional SMS phone verification.
- A Dholuo speech-to-text trial that the Luo developer (me) can check personally.
- Real service checks: send SMS login codes and ordinary test texts, receive real account emails, and receive push notifications on Android and iPhone. Keep a record of which tests passed and what paid usage they consumed.
- A tested plan for moving the website, database and media to the JOOUST server.
- Testing on Android and iPhone, store submissions, backup instructions and handover notes.

SERVICES AND PAYMENTS NEEDED

Cost estimates use KES 130 per US dollar. Final amount may change with exchange rates, tax and provider prices.

Service	What it is for	Estimated payment
Google Play Console	The account used to test and publish the Android app.	Assumed available through JOOUST. New registration: USD 25 once (~KES 3,250).
Apple Developer account	The account used to test and publish the iPhone app.	Assumed available via JOOUST. Apple membership USD 99/year (~KES 12,870). Fee waiver request possible.
Mac computer for iPhone release	Apple computer tools used to prepare and sign the iPhone app.	Assumed available through JOOUST, or online Mac service.
Africa's Talking account & SMS	Sends real text messages, including one-time sign-in codes and agreed urgent alerts.	JOOUST owns account. Entry plan KES 0.80/SMS before VAT (~KES 0.93 incl. VAT). 100 test messages ~KES 93.
Africa's Talking sender name	Registers the name shown on real SMS messages.	Budget KES 14,100 for Safaricom & Airtel together before VAT (~KES 16,356 with VAT). One network: KES 8,700 before VAT.
Africa's Talking short code	A short number that people can text for two-way SMS service.	Not needed for login codes or one-way alerts. Request separate quote only if two-way texting is required.

SERVICES AND PAYMENTS NEEDED (CONTINUED)

Service	What it is for	Estimated payment
Google Cloud Speech-to-Text	Changes spoken Dholuo into written words. Luo (Kenya) supported.	Test via developer key. Standard USD 0.016/min (~KES 2.08/min audio). Batch ~KES 0.39/min. JOOUST to setup production account.
Dholuo spoken audio / TTS	Reads Dholuo words aloud.	No computer voice confirmed. Use developer recordings for first release. Separate quote if computer voice required.
Email service (e.g., Resend)	Sends verification links, password resets and account notices.	JOOUST provides account. Resend free: 3,000/mo, 100/day. Paid plan USD 20/mo (~KES 2,600) if needed.
Firebase & Apple Push	Sends reminders and updates to phones.	KES 0 for basic notification services at current published prices.
Gemini	Runs current AI chat, search and summaries.	Charged according to use. Start with small monthly spending limit and monitor actual usage.
Vercel & Render	Free temporary places to run website and app connections during dev.	KES 0 within free limits. Not for public production host; Render services sleep after inactivity.
Neon database	Free online copy of database for building and testing.	KES 0 within free limits (up to 1 GB). Move live users to JOOUST server or paid production plan.
Cloudinary storage	Holds and displays pictures and videos during development.	KES 0 on free plan (25 monthly credits). Pay for plan or move files to JOOUST storage for public launch.
JOOUST server & domain	Hosts the public system after testing is complete.	JOOUST working server and domain. No new hosting/domain fee included assuming sufficient space and internet.

PAYMENT SUMMARY

- JOOUST is assumed to have a working Google Play account; otherwise arrange USD 25 (~KES 3,250) registration. iPhone publishing needs active Apple membership at USD 99/year (~KES 12,870) unless fee-waived.
- Use existing JOOUST domain; no new domain purchase included.
- Use Neon, Cloudinary, Vercel and Render free plans for development/testing within limits. JOOUST server is planned live host.
- JOOUST creates/owns Africa's Talking account, adds SMS credit (~KES 0.93 each incl. VAT) and gives test keys. Budget separately for sender-name registration.
- Africa's Talking sends SMS; email needs separate approved service. Gemini and speech-to-text costs depend on use.
- No payment collection service included as app is free. Add M-Pesa or store payment service if users will pay.
- Publishing on Google Play and Apple App Store makes apps available for download; stores do not host website, database or media.

CURRENT DATA AND AI READINESS

What is already available

- **Inspected Folder:** LUO DATA/Data/Data folder has 354 files (~326 MB): 119 Word docs, 208 pictures, 8 JSON files, 5 PDFs, 5 CSVs, 2 ZIPs, 2 Excels, 2 Markdowns, 1 TSV, 1 JSONL, 1 Python file.
- **Language Package:** Made from 26 transcripts, containing 2,708 text sections, 16 subject labels, 6 content-type labels, 88 exact repeat sections. Test split by interview/session: 17 training, 6 checking, 3 final test sessions. Marked research-only/internal.
- **Dholuo-English Table:** 2,708 rows, but English text is blank/unverified unless checked.
- **Media & Structure:** Pictures present, but no audio/video files and no labelled image-training collection. 354 files include research reports and working documents, not clean AI training datasets.

What each AI feature needs from the data

AI feature	What available data supports	What is still needed
Chatbot & Q&A	Gemini chat and search code exist. Approved records help find source material.	Choose approved public records, test real questions, check source citations, keep internal transcripts private.
Search & Culture	2,708 text sections and labels are a useful starting point.	Import permitted records, build search entries, test queries, confirm restricted records stay private.
Speech-to-Text	Google lists Luo (Kenya). Developer can make and check recordings.	No audio in folder. Make permission-cleared recordings and correct text; test words/speakers before claiming broad accuracy.
Text-to-Speech	Text can create a word/sentence list to record.	No ready-made Dholuo voice confirmed. Record audio or get quote for custom computer voice.
Translation & Grammar	Identifies Dholuo text with placeholder for English. Transcripts provide spoken examples.	Most translations blank. Add developer-checked English translations and clean written grammar examples.

> **Overall Data Readiness:** Useful for source-based chat, search, simple lessons and small experiments. Not ready to train dependable Luo translation, grammar, speech or image models without checked translations, paired audio, balanced examples and approvals.

TIMELINE

Estimated development time: ~15 weeks (7 hrs/day, 5 days/week = ~525 working hours). Phase One took ~10 days (~70 hours) for the web version.

Weeks	Main work
1–2	Confirm features, approved content, store accounts, SMS needs; prepare JOOUST server access.
3–5	Build Android and iPhone app foundations and connect to existing system.
6–8	Add learning content, search, lessons, quizzes, saved words and progress.
9–10	Add downloads, offline reading and progress syncing when internet returns.
11	Add email, optional SMS codes, phone notifications, AI chat and speech-to-text trial.
12	Connect tested system to JOOUST server, move database/media, check backups, prepare store submissions.
13–14	Test on real phones and with users, fix problems, complete store submissions.
15	Allow time for final fixes, interruptions, handover and store follow-up.

WHY PHASE TWO NEEDS ABOUT 15 WEEKS

- **Two phone systems:** Android and iPhone use different tools and store rules. Each needs separate setup and checks.
- **Offline use:** Lessons must download, stay available offline, and sync later without losing progress.
- **Service integration:** Email, SMS, push notifications, AI chat, speech-to-text and website must work together with failure checks.
- **Server migration:** Configure system on JOOUST server, move database/media, confirm access rules and test backups.
- **Extensive testing:** Real phone checks, poor internet simulations, user task testing, and bug fixing.
- **External factors:** Network/power interruptions, account verifications, sender-name approvals, and store review timelines.

ASSUMPTIONS AND ARRANGEMENTS

- **Accounts & Hosting:** JOOUST provides Google Play account, Apple Developer account/Mac access, working server/domain, and backups.
- **Cloud & Comms:** Neon, Cloudinary, Vercel, Render used for dev. JOOUST owns Africa’s Talking SMS account, adds credits, and provides email service access.
- **AI & Checks:** Developer uses personal account for Gemini/STT tests initially; production keys provided by JOOUST. Developer handles Luo language checks. No payment collection required.